



ZIMBABWE

**MINISTRY OF HIGHER AND TERTIARY EDUCATION,
SCIENCE AND TECHNOLOGY DEVELOPMENT**

**HIGHER EDUCATION EXAMINATIONS COUNCIL
(HEXCO)**

NATIONAL CERTIFICATE

IN

ELECTRICAL ENGINEERING

SUBJECT: Mathematics

PAPER CODE: 321/13/S04

DURATION: 3 Hours

OCTOBER/NOVEMBER 2015 EXAMINATION

REQUIREMENTS

Mathematical set, scientific calculator

INSTRUCTIONS TO CANDIDATE

- 1. Answer all questions.**
- 2. All questions carry equal marks.**
- 3. Geometrical construction to be done on a plain paper.**

QUESTION 1

- a) Construct a 45° angle. (2 marks)
- b) Copy the 45° angle, on it, complete a right angled isosceles triangle whose base is 100mm. (5 marks)
- c) Inscribe the triangle in (b). (3 marks)

QUESTION 2

- a) Rationale $\frac{(2-3\sqrt{2})(5+2\sqrt{2})}{4+4\sqrt{2}}$ (4 marks)
- b) Simplify $(4^{4m} \times 2^{2n}) \div (16^m \times 8^{2n})$ (3 marks)
- c) Solve $9^{(x-1)} = 27^{(x-1)}$ (3 marks)

QUESTION 3

- a) Solve the simultaneous equations:
 $2x + 2y = 9$ and $2x - y = 9$ (3 marks)
- b) For $y = 6 + x - x^2$ determine
 i) y -intercept (1 mark)
 ii) x -Intercept (3 marks)
 iii) Turning point (3 marks)

QUESTION 4

- a) If θ is small, give an approximation for
 $\frac{\theta \sin \theta}{\theta^2}$ (2 marks)
- b) Express $\sin 7\theta + \sin 5\theta$ as a product of sines and cosines. (2 marks)
- c) An electric current is given by $i = 200 \sin 300t + 100 \cos 300t$. Express this current in the form
 $R \sin (wt \pm a)$ (6 marks)

$$\sin A + \sin B = 2 \sin \left(\frac{A+B}{2} \right) \cos \left(\frac{A-B}{2} \right)$$

$$\frac{200}{2} \sin 300t + \frac{100}{2} \cos 300t = \frac{b}{x}$$

QUESTION 5

a) Differentiate with respect to x

i) $y = \cos 3x$

(1 mark)

ii) $y = (x^2 - 5x + 3)^5$

(4 marks)

b) Differentiate $y = 2x^2 + 2$ from first principles.

(5 marks)

QUESTION 6

a) Evaluate $\int \frac{(3x^2+2)}{x^3+2x+7} dx$

(5 marks)

b) Determine the area enclosed by the curve $y = 2x + 3$, the x axis, the ordinates $x = 1$ and $x = 4$.

(5 marks)

QUESTION 7

a) The two vectors are $V_1 = i + 2j$; and $V_2 = 3i - 5j$.

Determine:

i) The dot product $(V_1 \cdot V_2)$

(2 marks)

ii) The angle between the vectors.

(3 marks)

b) A circuit has two impedances of $(5 + j5)$ ohms and $(6 - j8)$ ohms connected in parallel. Determine the total impedance of the circuit in the form $a + jb$.

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QUESTION 8

An object follows a path OX given by $x = t^3 - 6t^2 + 9t$ metres.

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a) At what time and in what position does the object attain zero velocity.

(5 marks)

b) What will be the acceleration at those instances?

(3 marks)

c) Determine the velocity when the acceleration is zero.

(2 marks)

QUESTION 9

- a) Convert to binary
- i) FAC_{16} (2 marks)
 - ii) 357_8 (2 marks)
- b) Convert $110, 011_2$ to decimal. (2 marks)
- c) A record rotates at $100/3$ revolutions per minute. Determine the angular velocity of the record in radians per second. (4 marks)

Q QUESTION 10

- a) A load is pulled by a rope which makes 45° with the ground. If the tension of the rope is 60 N, Calculate the work done if the load is pulled for 2 km. (5 marks)
- b) An electric motor is developing 8 kw at a speed of 1200 rpm. Calculate the torque developed by the motor. (5 marks)

.../bc

$$1200 \times 60 \times 2$$

$$P_{out} = P_{in} \cos \theta$$
$$W = S \cdot \omega$$